Ike’s FUTURE HOUSE

1. Design Choices

The design philosophy for **Ike’s FUTURE HOUSE** was centered on creating a high-end, modern architectural visualization that balances aesthetic luxury with user-friendly interactivity.

- **Scene Composition:** I chose a "Contemporary Estate" theme. The centerpiece is a high-fidelity GLTF house model, complemented by a curated selection of luxury assets, including a McLaren, a Range Rover, and professional landscaping elements like the pool and parasols.
- **User Interface (The Tour Menu):** To solve the common VR issue of "getting lost" in a large scene, I designed a custom **Teleportation UI**. Using a fixed HTML overlay and JavaScript functions, users can instantly move between the Ground Floor, First Floor, and Backyard, ensuring they see every part of the design.
- **Atmospheric Lighting:** I implemented a multi-layered lighting system. Ambient light provides a baseline visibility, while specific **Spotlights** and **Point lights** were placed to mimic architectural lighting on the gate and streetlights, creating depth and shadows.
- **Water Simulation:** For the pool, I utilized the ada-water.js component rather than a simple flat plane. This adds transparency and light refraction, which are essential for a realistic luxury backyard feel.

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2. Technical Challenges & Solutions

Developing a functional 3D environment within a browser environment presented several technical hurdles:

- **Asset Management and Scaling:**
  - *Challenge:* External GLTF models often have different "world scales." When imported, the McLaren was initially larger than the house.
  - *Solution:* I used the A-Frame Inspector and iterative manual adjustments to normalize scales (e.g., scaling the fence to 0.065 0.0085 0.015) so all components looked cohesive.
- **Navigation and Camera Control:**

- *Challenge:* Standard WASD controls can be slow for traversing a large estate, and users can accidentally "fly" away from the scene.
- *Solution:* I modified the camera entity to include movement-controls="fly: true" for freedom of movement but anchored the experience with the **JavaScript tp() function**. This allows the camera's position attribute to be updated instantly via button clicks.

- **Code Optimization:**
  - *Challenge:* Loading high-poly models can cause "jank" or lag during the initial render.
  - *Solution:* I utilized the <a-assets> tag to ensure all 3D models and textures are pre-cached by the browser before the scene is displayed to the user.

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3. Future Improvements

While the current version meets all examination requirements, I have identified several areas for future iteration:

- **Physics and Collisions:** I plan to integrate the aframe-physics-system to add solid boundaries to walls and floors, preventing the user from walking through solid objects.
- **Interactive Event Listeners:** I would like to add "Click-to-Open" functionality for the front gate and car doors using A-Frame's animation component triggered by a cursor component.
- **Environmental Audio:** Adding a sound component with "spatial audio" would allow the user to hear the pool water splashing or the hum of the streetlights as they walk past them, significantly increasing immersion.
- **Dynamic Skybox:** Implementing a script to transition the a-sky texture from day to night based on a timer, synced with the switching on of the interior point lights.